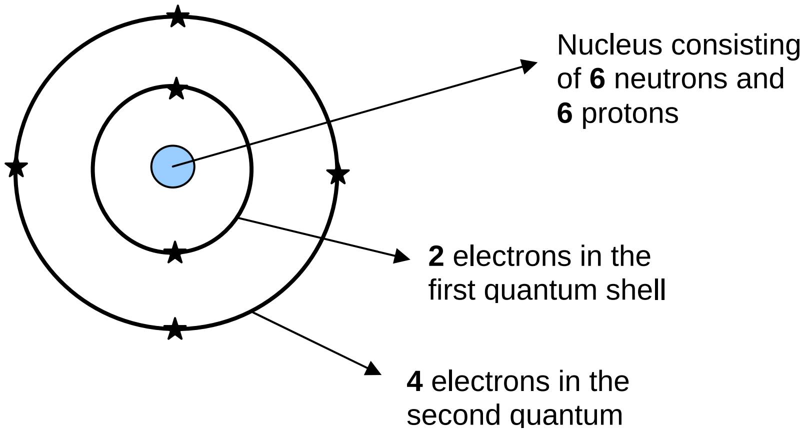
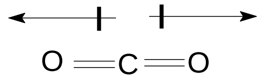
2010 AJC H1 Chemistry Preliminary Exam_ Answers

1 (a)



(b) (i)



Linear molecule

Dipoles of the C=O bonds cancel each other

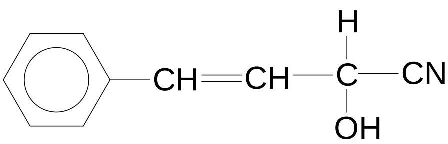
(ii) Electronegativity: O > S > Se

Polarity: C=O > C=S > C=Se

(c) (i) $Fe^{2+} + C_2O_4^{2-} \rightarrow Fe^{3+} + 2CO_2 + 3\ e^-$

(ii) Vol of $KMnO_4$ = 15 cm^3

2 (a) (i)



(ii)

Stage	Reagent(s) and Condition	Type of reaction
I	HCN, NaCN / trace NaOH	Addition
II	dil. H_2SO_4 / HCl, Heat (under reflux)	Hydrolysis

(b) H_2, Ni catalyst at 140^oC or

H_2, Pt catalyst at r.t.p.

(c) (i) $ROH + PCl_5 \rightarrow RCl + POCl_3 + HCl$

White fumes (of HCl) will be observed.

(ii) $\Delta H = (+1204) - (+1600) = -396\ kJ\ mol^{-1}$

(iii) ΔH_1 = 4 × enthalpy change of formation of PCl_3(g)

$\Delta H_1 + \Delta H = \Delta H_2$

3 **(a)** **(i)** $MgCl_2$: any value between 6 ~ 7

$SiCl_4$: any value between 0 ~ 2

(ii) $MgCl_2$ is ionic,

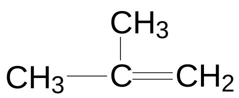
Undergoes slight hydrolysis / no hydrolysis

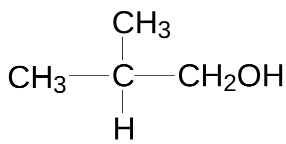
$SiCl_4$ is covalent,

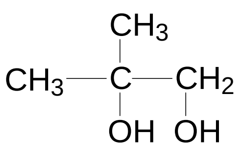
undergoes hydrolysis readily in water,

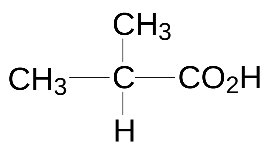
$SiCl_4 + 2H_2O \rightarrow SiO_2 + 4HCl$

(b)

C = 

D = 

E = 

F = 

(c) **(i)** Structural isomerism

(ii) 2–chlorobutane destroys the ozone layer.

4 **(a)** **(i)** order of reaction wrt **G** is **<u>1</u>**.

order of reaction wrt NaOH is **<u>1</u>**.

(ii) rate = k [**G**] [NaOH]

(iii) k = 0.0556 $mol^{-1}\ dm^3\ min^{-1}$

(b) **(i)** rate = k [**H**]

(ii) Different; rate law for reactions are different.

(c) **(i)**

Stage	Change in hybridisation of carbon
I	$sp^3 \rightarrow sp^2$
II	$sp^2 \rightarrow sp^3$

(ii)  sp^3 hybrid

(d) **(i)**

III	KCN in ethanol, heat under reflux
IV	Li*Al*H_4 in dry ether, room temperature (or 0^oC)

(ii) reduction

5 **(a)** **(i)** $P_4S_3 + 8O_2 \rightarrow 3SO_2 + P_4O_{10}$

(ii) $SO_2 + H_2O \rightarrow H_2SO_3$ (pH < 2)

$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$ (pH < 2)

(iii) Both SO_2 and P_4O_{10} have simple molecular structure with weak van der Waals forces between the molecules SiO_2 has a giant molecular structure with strong, extensive covalent bonds between Si and O atoms.

Hence SO_2 and P_4O_{10} have m.p. that are similar but much lower than that of SiO_2.

(b) **(i)** $K_c = \dfrac{[SO_3]^2}{[SO_2]^2[O_2]}$

$= 369\ mol^{-1}\ dm^3$

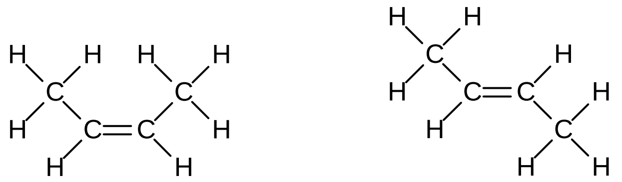
(ii) % of SO_2 in mixture = 1.62 %

(iii) $[SO_3]$ in the sample = $0.0344\ mol\ dm^{-3}$

(iv) temperature decrease
position of equilibrium will shift to the right
and rate of achieving equilibrium will decrease

(c) **(i)** **II** : excess concentrated H_2SO_4, 170 / 180 ^oC

(ii)



cis-but-2-ene trans-but-2-ene

There will be bond strain (or angle strain) for the trans cyclobutene.

6 **(a)** **(i)** $2\ NaOH + Al_2O_3 + 3H_2O \rightarrow 2NaAl(OH)_4$

MgO is basic, Al_2O_3 is amphoteric.

Hence only Al_2O_3 will react with NaOH.

(ii) Mg^{2+} has a higher ionic charge than Na^+ , Mg^{2+} is smaller than Na^+

Stronger electrostatic forces of attraction between ions in MgO, hence L.E. (MgO) numerically larger than L.E. (Na_2O)

(b)

1. **J** undergoes hydrolysis when refluxed with aq NaOH, followed by acidification to form **K** and **L**.
J = ester

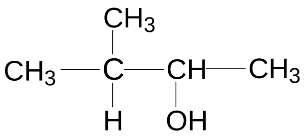
2. **K** gives positive triiodomethane test as yellow CHI_3 ppt is formed
$\Rightarrow$ CH_3CO- or $CH_3CH(OH)$- group present

3. **K** undergoes elimination of H_2O (dehydration) to give alkene **M** / **K** is alcohol, $CH_3CH(OH)$- group

4. **M** undergoes oxidative cleavage (oxidation with breaking of double bond) when heated with acidified $KMnO_4$, **M** is an alkene

5. Since propanone is produced, **M** is $(CH_3)_2C=CHCH_3$

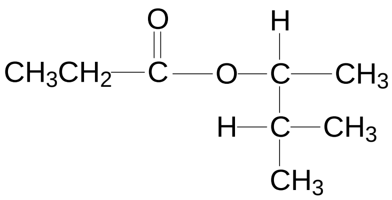
6.

K = 

7. **L** reacts with carbonate to liberate CO_2

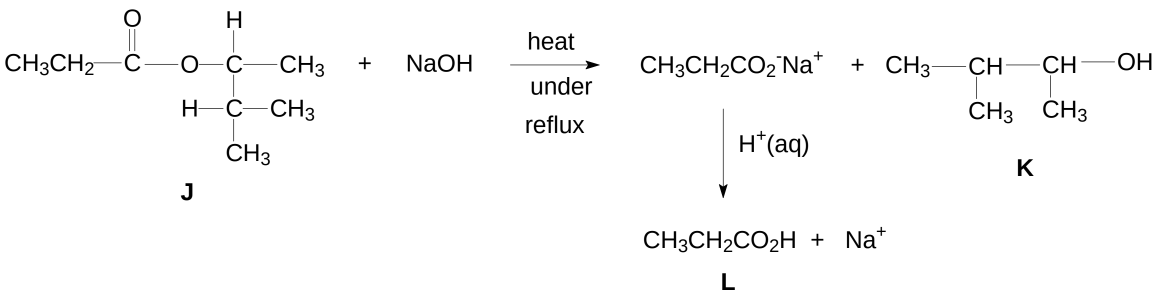
8. **L** is a propanoic acid, $CH_3CH_2CO_2H$

9.

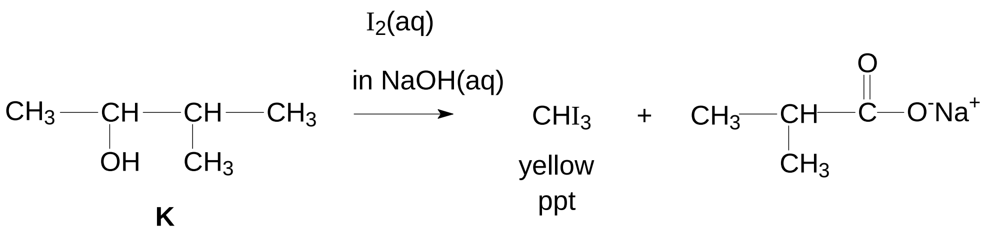
J = 

Equations:

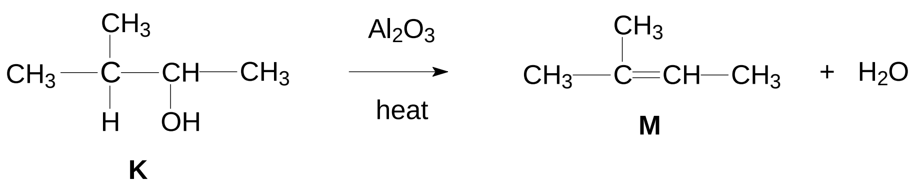
(1) Hydrolysis of ester **J**



(2) Triiodomethane test



(3) Dehydration of K



(4) Oxidative cleavage of M

$$CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}=CHCH_3 \quad + \quad 3[O] \longrightarrow CH_3COCH_3 \quad + \quad CH_3CO_2H$$

M — propanone

(5) Reaction of acid L with sodium carbonate

$$2CH_3CH_2CO_2H \; + \; Na_2CO_3 \longrightarrow 2CH_3CH_2CO_2Na \; + \; CO_2 \; + \; H_2O$$

(c) **(i)** Standard enthalpy change of combustion is the amount of heat evolved when one mole of **M**, C_5H_{10}, is completely burnt in excess oxygen at 1 atm, 25^oC.

(ii) $C_5H_{10} + 15/2O_2 \rightarrow 5CO_2 + 5H_2O$

Volume of air required = 375 cm^3

7 **(a)** **(i)** Strong acids completely dissociate in water to produce H_3O^+ / H^+.
Weak acids undergo partial dissociation in water to produce H_3O^+.

(ii) $[H^+] = 10^{-2.60} = 2.51 \times 10^{-3}$ mol dm^{-3}
<< [benzoic acid] = 0.1 mol dm^{-3}
$\therefore$ benzoic acid is a weak acid.

(b) **(i)** pH = 13.2

(ii) volume of potassium hydroxide required
= 13.3 cm^3

(iii)
- Weak acid – strong base titration, so pH of at the equivalence point > 7
- Equivalence point lies within the working pH range of phenolphthalein indicator.
- Phenolphthalein gives a distinct colour change from colourless to pink.

(c) On addition of H^+,
$C_6H_5COO^- + H^+ \rightarrow C_6H_5COOH$

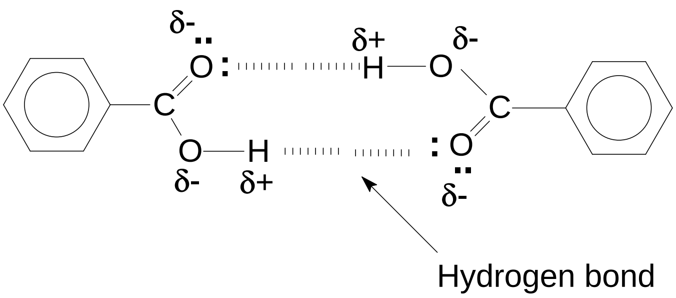
On addition of OH^-,
$C_6H_5COOH + OH^- \rightarrow C_6H_5COO^- + H_2O$

Hence, pH remains relatively constant

(d) Enthalpy change of neutralisation of KOH by benzoic acid is **less exothermic** than -57.3 kJ mol^{-1}.

Heat is absorbed to completely dissociate the benzoic acid into the H^+ and $C_6H_5COO^-$ ions.

(e) **(i)**



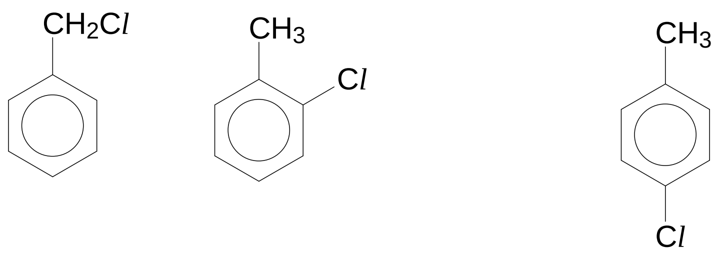
(ii) Blue litmus remains blue / no change in colour of blue litmus / blue litmus does not turn red.

Benzoic acid does not dissociate into H^+ and $C_6H_5CO_2^-$ in non-polor benzene solvent. Hence, solution is not acidic.

(f) **(i)** $KMnO_4$(aq), H_2SO_4(aq), heat (under reflux)

(ii) $C_6H_5CH_3 + 3[O] \rightarrow C_6H_5COOH + H_2O$

(g) **(i)**



P **Q** **R**

(ii) **P** : UV light / high temp

Q, **R**, : $FeCl_3$ or $AlCl_3$

(iii) Methylbenzene has a <u>lower</u> boiling point than **P** / **Q** / **R**

Van der waals forces of attraction in methylbenzene is weaker than dipole-dipole forces in **P** / **Q** / **R**